

$\rightarrow L^{\infty}$

→ optimal

7.

$\times$

7

2
4
0
3
✓

$$\rightarrow M$$

→ LRU (Least-Recently used)

7	0	1	2	0	3	0	4	2	3	0
			2	2	2	2	2	2	2	2
		1	1	1	1	1	4	4	4	4
	0	0	0	0	0	0	0	0	0	0
7	7	7	7	7	3	3	3	3	3	3
x	x	x	x	✓	x	✓	x	✓	✓	✓
9	0	3	2	1	2	0	1	7	0	1
2	2	2	2	2	2	2	2	2	2	2
4	4	4	4	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0
4	3	3	3	3	3	3	3	3	3	3
3	✓	✓	✓	✓	x	✓	✓	x	✓	✓
x										

Hit-Ratio = $\frac{14}{22}$

→ Miss Ratio = $\frac{8}{22}$

→ MRU (Most Recently Used)

	7	0	1	2	0	3	0	4	2	3	0
				2	2	2	2	2	2	3	0
			1	1	1	1	1	1	1	1	0
		0	0	0	0	3	0	4	4	4	1
7	7	7	7	7	7	7	7	7	7	7	7
X	X	X	X	✓	X	X	X	✓	X	X	X
3	0	3	2	1	2	0	1	7	0	0	1
3	0	3	2	2	2	0	0	0	0	0	0
4	1	1	1	1	1	1	1	1	1	1	1
4	4	4	4	4	4	4	4	4	4	4	4
1	7	7	7	7	7	7	7	7	7	7	7
X	X	X	X	✓	✓	X	✓	✓	✓	✓	✓

Hit = $\frac{8}{22}$, mis = $\frac{14}{22}$

→ optimal (impotent) → Prediction of future page sequence

	7	0	1	2	0	3	0	4	2	3	0	3	0	3
				2	2	2	2	2	2	2	2	2	2	2
			1	1	1	1	1	4	4	4	4	4	4	4
		0	0	0	0	0	0	0	0	0	0	0	0	0
7	7	7	7	7	7	3	3	3	3	3	3	3	3	3
X	X	X	X	✓	X	✓	X	✓	✓	✓	✓	✓	✓	✓

	2	1	2	0	1	7	0	1
2	2	2	2	2	2	7	7	7
4	4	4	4	4	4	4	4	4
0	0	0	0	0	0	0	0	0
3	1	1	1	1	1	1	1	1
✓	X	✓	✓	✓	X	✓	✓	✓

Hit = $\frac{14}{22}$

Miss = $\frac{8}{22}$

Sequence for the F
- miss
- optim

Phys

Sequence

3 2 1

→ FIFO

		2	
3	3		
X	X		

Hit =

→ LRU

		2	2
3	3		
X	X		

Hit

Sequence of pages. Find Hit, Miss Ratio using FIFO, LRU, MRU and optimal technique.

Physical memory = 3.

Sequence :-

3 2 1 3 4 1 6 2 4 3 4 2 1 4 5 2 1 3 4 - 19

→ FIFO

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
Hit		1	1	1	1	2	2	2	2	2	2	1	1	1	1	1	1	3	3
Miss	2	2	2	2	2	6	6	6	6	4	4	4	4	4	4	2	2	2	2
Count	3	3	3	3	4	4	4	4	4	3	3	3	3	3	5	5	5	5	4
Result	X	X	X	✓	X	✓	X	X	✓	X	X	✓	X	✓	X	X	✓	X	X

Hit = $\frac{6}{19}$ Ratio = $\frac{13}{19}$ Miss

⇒ LRU

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
Hit		1	1	1	1	1	1	2	2	2	2	2	2	2	5	5	5	3	3
Miss	2	2	2	4	4	4	4	4	4	3	3	3	1	1	1	2	2	2	4
Count	3	3	3	3	3	3	6	6	6	6	4	4	4	4	4	4	4	1	1
Result	X	X	X	✓	X	✓	X	X	✓	X	X	✓	X	✓	X	✓	X	X	X

Hit = $\frac{5}{19}$ Miss = $\frac{14}{19}$

=> MRU

3	2	1	3	4	1	6	2	4	3	4	1	4	5	2	1	3	4
		1	1	1	1	6	6	6	6	6	6	6	6	6	6	6	6
2	2	2	2	2	2	2	2	2	2	2	2	1	1	1	1	1	3
3	3	3	3	4	4	4	4	4	3	4	4	4	4	5	2	2	2
X	X	X	✓	X	✓	X	✓	✓	X	X	✓	X	✓	X	X	✓	X

Hit = $\frac{7}{19}$ miss = $\frac{12}{19}$

=> OPTimal

3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
		1	1	1	1	6	6	6	3	3	3	1	1	1	1	1	1	4
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3
3	3	3	3	4	4	4	4	4	4	4	4	4	4	4	5	5	5	5
X	X	X	✓	X	✓	X	✓	✓	X	✓	✓	X	✓	X	✓	✓	X	X

Hit = $\frac{9}{19}$ miss = $\frac{10}{19}$

-> reference checking

Apply Fifo Page replacement algorithm for the following seq. of Pages, check whether Belady's anomaly exist or not.

-> To check BA take phy mem size = 3 or 4 frames.

1 2 3 4 1 2 5 1 2 3 4 5

-> FIFO 3 Frame

1	2	3	4	1	2	5	1	2	3	4	5
		3	3	3	2	2	2	2	2	4	4
	2	2	2	1	1	1	1	1	3	3	3
1	1	1	4	4	4	5	5	5	5	5	5
X	X	X	X	X	X	X	✓	✓	X	X	✓

So

$$\text{Faults} = 10 = \frac{10}{12}$$

$$\text{Hit} = 2 = \frac{2}{12}$$

	1	2	3	4	1	2	5	1	2	3	4	5
			3	3	3	2	2	2	2	2	2	5
	2	2	2	2	1	1	1	1	1	1	4	4
	1	1	1	4	4	4	5	5	5	3	3	3
X	X	X	X	X	X	X	X	✓	✓	X	X	X

$$\text{Hit} = \frac{2}{12}, \text{ miss} = \frac{10}{12}$$

For 4 frames FIFO

	1	2	3	4	1	2	5	1	2	3	4	5
			3	4	4	4	4	4	4	3	3	3
			2	3	3	3	3	3	2	2	2	2
			2	2	2	2	2	1	1	1	1	5
			1	1	1	5	5	5	5	5	4	4
X	X	X	X	X	✓	✓	X	X	X	X	X	X

$$\text{Hit} = \frac{2}{12}, \text{ miss} = \frac{10}{12}$$

∴ Belady's Anomaly exist for FIFO because

for 3 frames $\rightarrow$ Hit Ratio is $\frac{3}{12}$

for 4 frames $\rightarrow$ Hit Ratio is $\frac{2}{12}$

So, increasing frames increases faults.